

Jasper[®] CDC

NYCU-EE IC LAB SPRING 2024

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Outline

- Structural Analysis
- Functional Checks

Structural Analysis

Check the circuit structure

In our tcl scripts

```
# Find Clock Domains
check_cdc -clock_domain -find
# Find CDC pairs
check_cdc -pair -find
# Find Schemes
check_cdc -scheme -add handshake -module Handshake_syn -map
    {{data din} {sreq sreq} {dreq dreq} {dack dack} {sack sack}}
check_cdc -scheme -add fifo -module FIFO_syn -map
    {{rdata rdata} {wdata wdata} {wptr wptr} {rptr rptr} {wfull wfull} {rempty rempty} {winc winc} {rinc rinc}}
check_cdc -scheme -find
# Find Convergence
check_cdc -group -find
```

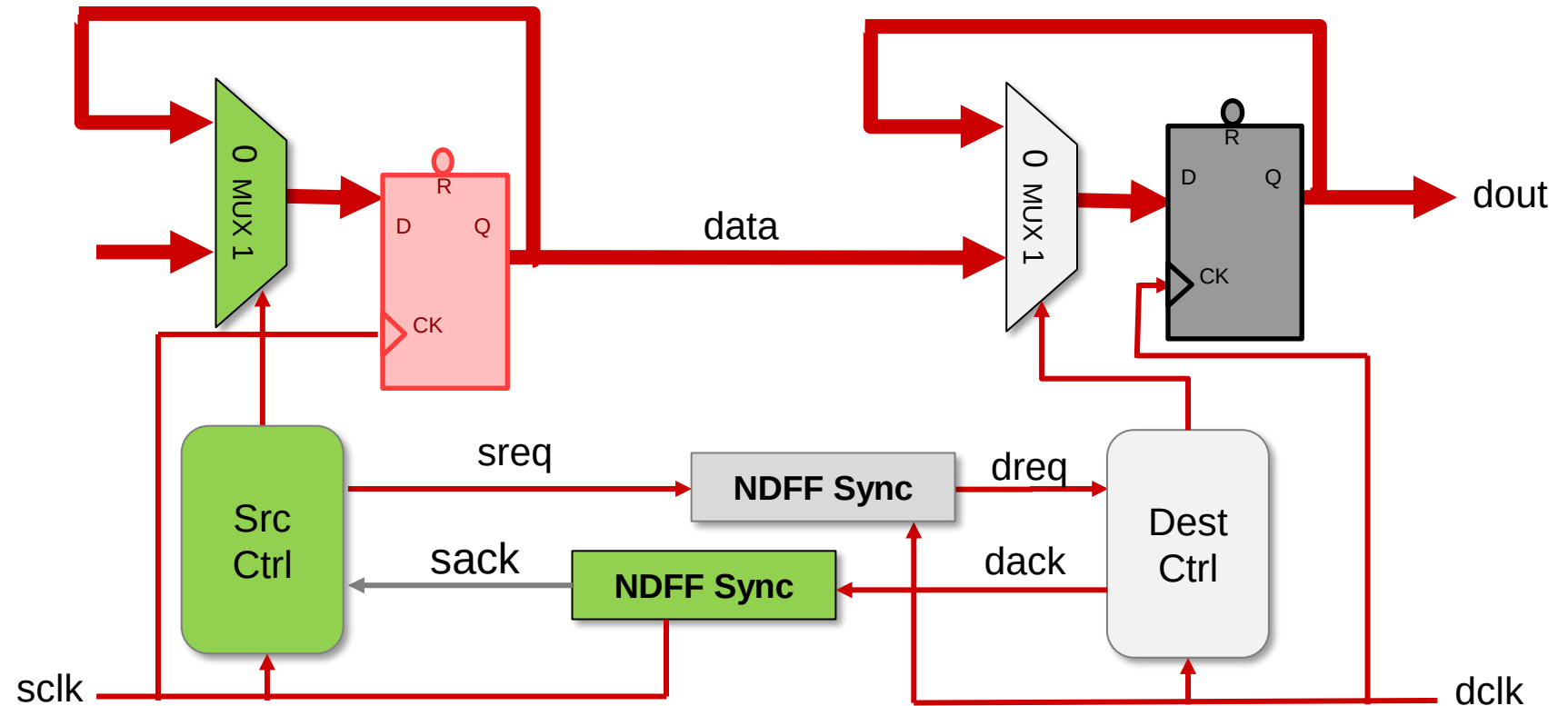
Find Schemes

- In this lab, there are two synchronizers you need to design.
 - Handshake
 - FIFO
- Please use the synchronizer we designed to complete your synchronizer.
 - Use NDFF_syn in Handshake to transfer req / ack signal.
 - Use NDFF_BUS_syn in FIFO to transfer gray-coded pointers.
- We have already map the module name to the pre-defined schemes
 - DO NOT change the module name.
- We have already map the signal to the formal signal
 - DO NOT change the signal name.

Handshake

```
check_cdc -scheme -add handshake -module Handshake_syn -map
```

```
{{data din} {sreq sreq} {dreq dreq} {dack dack} {sack sack}}
```

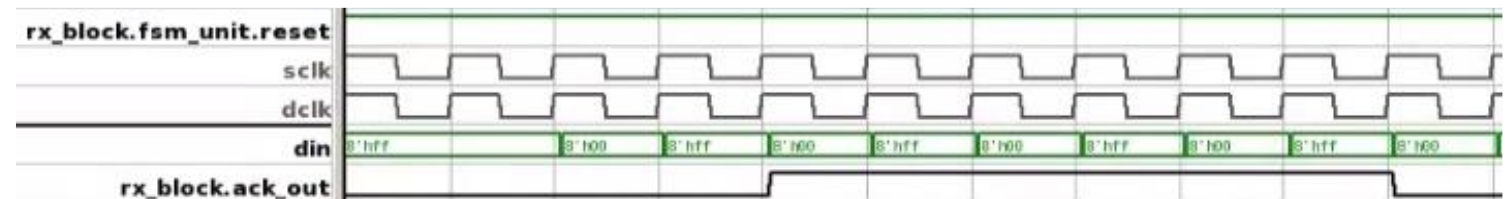


Handshake

- We have design the NDFF synchronizer for you.
- This step checks the following structures.
 - **ACK_WO_SREQ**: This check indicates that the acknowledgment signal in the destination domain is de-asserted before the request signal is de-asserted in the source domain.



- **DAT_HS_STBL**: This check indicates that the data from the sender becomes unstable before it receives an acknowledgment from the receiver.

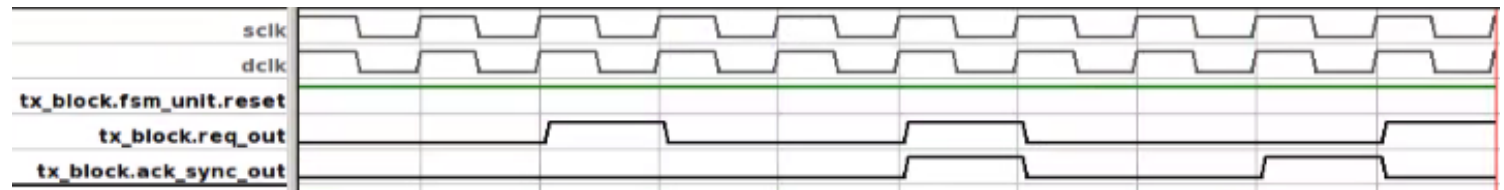


Handshake

- **NAK_WO_SREQ**: This check indicates that the receiver asserts a new acknowledgment before a new request is received.



- **NRQ_WO_DACK**: This check indicates that the sender submits a new request before the acknowledgment for the previous transfer is de-asserted.



Handshake

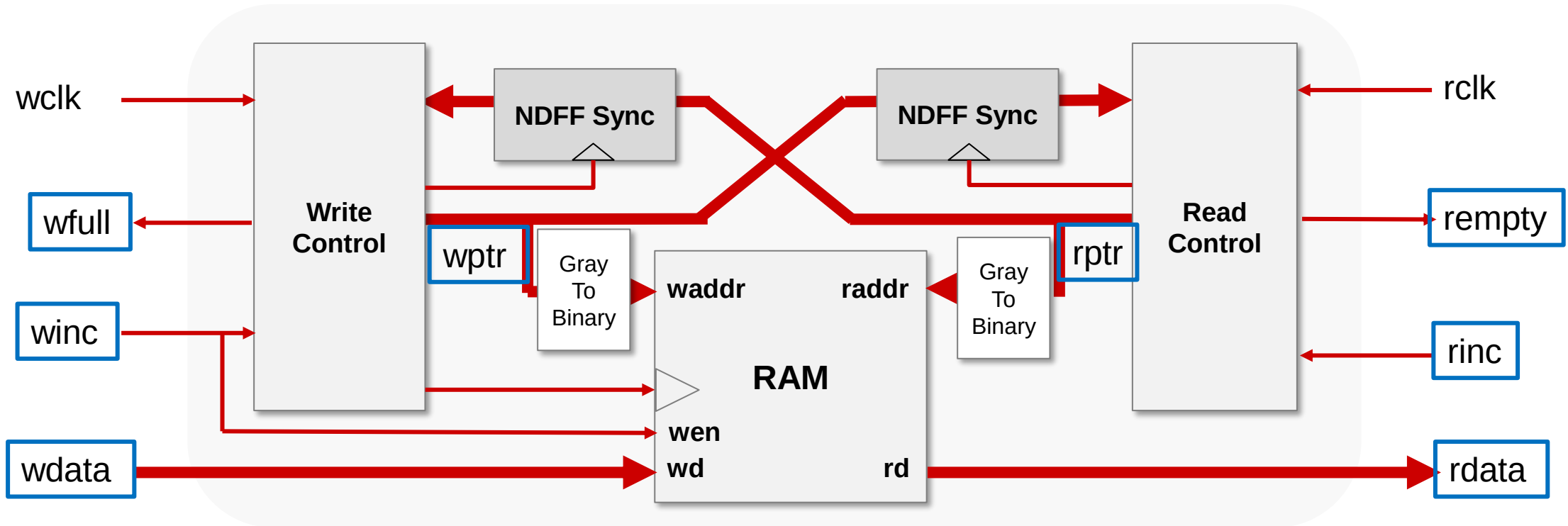
- **REQ_NO_HOLD**: This check indicates that the request signal is de-asserted before receiving acknowledgment from the destination clock domain.



FIFO

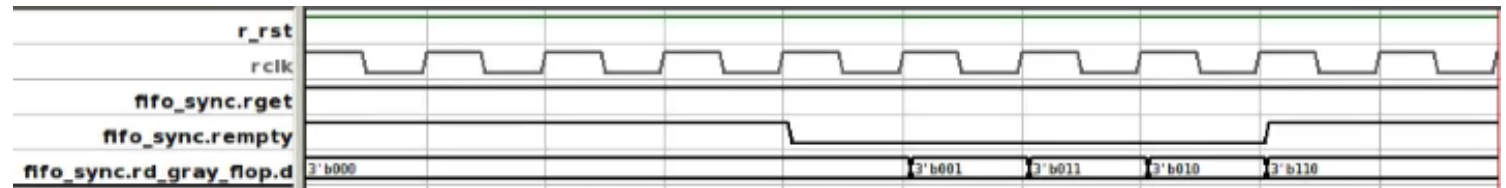
```
check_cdc -scheme -add fifo -module FIFO_syn -map
```

```
{{rdata rdata} {wdata wdata} {wptr wptr} {rptr rptr} {wfull wfull} {rempty rempty}  
{winc winc} {rinc rinc}}
```



FIFO

- We have design the NDFF_BUS synchronizer for you.
- This step checks the following structures.
 - **POP_ON_EMPTY**: This check indicates that it is possible to read from an empty FIFO.



- **PSH_ON_FULL**: This check indicates that it is possible to write to a full FIFO.

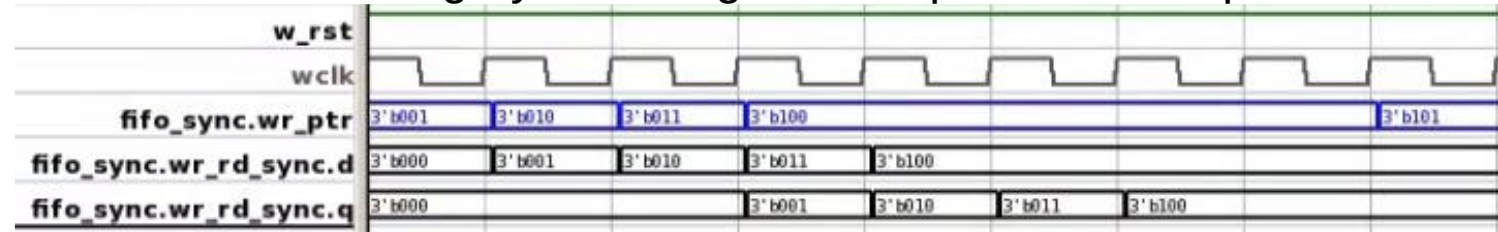


FIFO

- **RPT_NO_GRAY**: This check indicates that the gray encoding for the specified read pointer of the specified FIFO has failed.



- **WPT_NO_GRAY**: This check indicates that the gray encoding for the specified write pointer of the specified FIFO has failed.



- You should add one more stage of the DFF after the read data output by our dual port SRAM to make sure the data synchronize to read clock domain.
 - Or you will get CDC_NO_SYNC violation

After Structure Analysis

- You may have problems with pairs / schemes / convergence.

CDC Phases

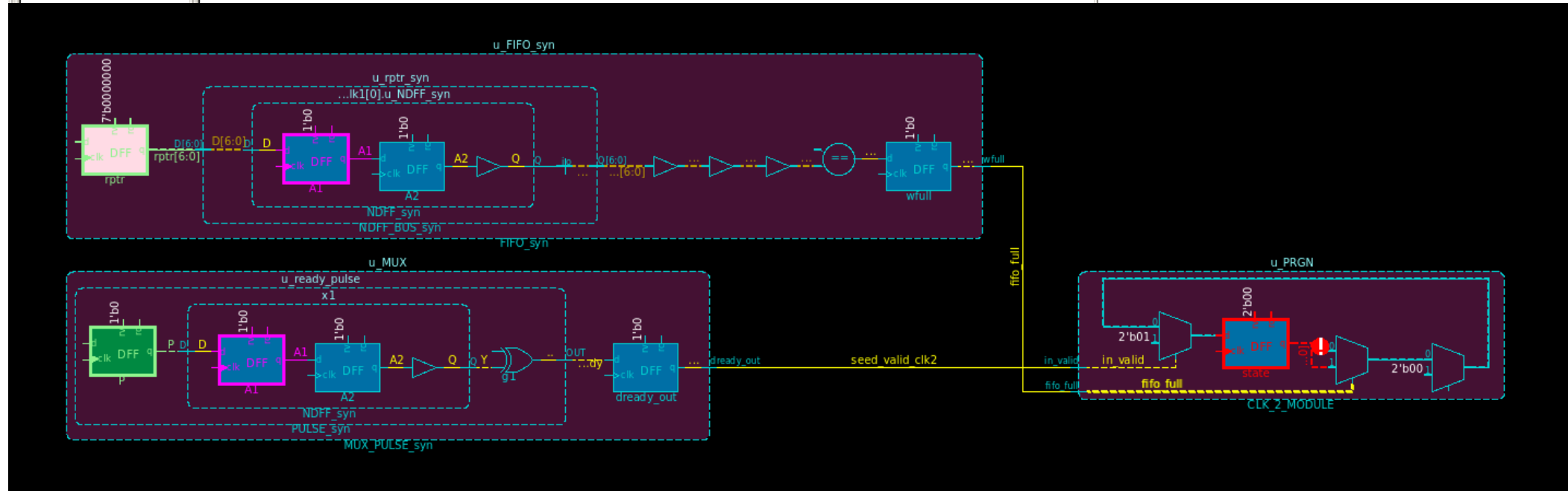
Pairs Schemes Convergence Functional Metastability

All

Convergence

	CDC Group	Type	Same Source Domain	Same Source Signal	FSM	Depth
✗	u_PRGN.state	Convergence	False	False	False	0
✗	u_PRGN.out_cnt	Convergence	False	False	False	0

Double click



After Structure Analysis

- You should pass all check for pairs / schemes / convergence.

The image displays three screenshots of the CDC Phases tool, demonstrating successful checks for Pairs, Schemes, and Convergence.

Top Screenshot: Pairs Check

The 'Pairs' tab is selected. The left pane shows a tree view of clock pairs: clk2 -> clk3, clk1 -> clk2, clk3 -> clk2, and clk2 -> clk1. The main table lists the following pairs, all with green checkmarks:

Source Unit	Destination Unit	Sou	De	Lo	Loc	Pair	Tag	Scheme
clk2	u_FIFO_syn.rdata	clk2	clk3	clk2	clk3	Data		u_FIFO_syn.rdata
u_FIFO_syn.rptr[0]	...yn.genblk1[0].u_NDFF_syn.A1	clk3	clk2	clk3	clk2	Control		u_FIFO_syn.rdata
u_FIFO_syn.rptr[1]	...yn.genblk1[1].u_NDFF_syn.A1	clk3	clk2	clk3	clk2	Control		u_FIFO_syn.rdata
u_FIFO_syn.rptr[2]	...yn.genblk1[2].u_NDFF_syn.A1	clk3	clk2	clk3	clk2	Control		u_FIFO_syn.rdata
u_FIFO_syn.rptr[3]	...yn.genblk1[3].u_NDFF_syn.A1	clk3	clk2	clk3	clk2	Control		u_FIFO_syn.rdata
u_FIFO_syn.rptr[4]	...yn.genblk1[4].u_NDFF_syn.A1	clk3	clk2	clk3	clk2	Control		u_FIFO_syn.rdata
u_FIFO_syn.rptr[5]	...yn.genblk1[5].u_NDFF_syn.A1	clk3	clk2	clk3	clk2	Control		u_FIFO_syn.rdata
u_FIFO_syn.rptr[6]	...yn.genblk1[6].u_NDFF_syn.A1	clk3	clk2	clk3	clk2	Control		u_FIFO_syn.rdata
u_FIFO_syn.waddr[5:0]	u_FIFO_syn.rdata	clk2	clk3	clk2	clk3	Data		u_FIFO_syn.rdata
u_FIFO_syn.wfull	u_FIFO_syn.rdata	clk2	clk3	clk2	clk3	Data		u_FIFO_syn.rdata
u_FIFO_syn.wptr[0]	...yn.genblk1[0].u_NDFF_syn.A1	clk2	clk3	clk2	clk3	Control		u_FIFO_syn.rdata
u_FIFO_syn.wptr[1]	...yn.genblk1[1].u_NDFF_syn.A1	clk2	clk3	clk2	clk3	Control		u_FIFO_syn.rdata
u_FIFO_syn.wptr[2]	...yn.genblk1[2].u_NDFF_syn.A1	clk2	clk3	clk2	clk3	Control		u_FIFO_syn.rdata

Bottom Left Screenshot: Schemes Check

The 'Schemes' tab is selected. The table shows the following schemes, all with green checkmarks:

Scheme	Scheme Type	Detection Type	Tag
u_Handshake_syn.din	Handshake	UserDefined	
u_FIFO_syn.rdata	FIFO	UserDefined	

Bottom Right Screenshot: Convergence Check

The 'Convergence' tab is selected. The table shows the following convergence checks, all with green checkmarks:

Source Unit	Destination Unit	Source Clock Domain	Destination Clock Domain	Local Source Clock	Local Destination Clock	Pair Class
clk2	clk3	clk2	clk3	clk2	clk3	
clk1	clk2	clk1	clk2	clk1	clk2	
clk3	clk2	clk3	clk2	clk3	clk2	
clk2	clk1	clk2	clk1	clk2	clk1	

Functional Analysis

Check the usage of the synchronizers

In our tcl scripts

```
## ----- Functional Checks ----- ##  
check_cdc -protocol_check -generate  
check_cdc -protocol_check -prove
```


Functional Analysis

- In this lab, there are two synchronizers you need to design.
 - Handshake
 - MUX_PULSE
 - FIFO
- You must abide by these synchronizer usage rules.
- In this step, JG will use formal verification to prove your circuit follows the rule (pre-defined assertion) we introduce in structure analysis.
- You will learn the formal verification in the later lab, at that time you will write your own assertion and use JG to prove it.

Handshake

- **ACK_WO_SREQ:**

- `@(posedge u_Handshake_syn.dclk) disable iff (~u_Handshake_syn.rst_n) (u_Handshake_syn.dack) && (u_Handshake_syn.dreq) | => (u_Handshake_syn.dack)`
- At every positive edge of the dclk clock signal, if the reset RST_N is high, then if the dack and dreq is high at the destination domain, dack must be high at the next cycle.

- **DAT_HS_STBL:**

- `@(posedge u_Handshake_syn.dclk) disable iff (~u_Handshake_syn.rst_n) (u_Handshake_syn.dreq && !(u_Handshake_syn.dack)) | => $stable(u_Handshake_syn.din)`
- At every positive edge of the dclk clock signal, if the reset RST_N is high, and if the dreq is high and dack is low at the destination domain, then din must be stable at the next cycle.

Handshake

- **NAK_WO_SREQ:**

- `@(posedge u_Handshake_syn.dclk) disable iff (~u_Handshake_syn.rst_n) !(u_Handshake_syn.dack) && !(u_Handshake_syn.dreq) | => !(u_Handshake_syn.dack)`
- At every positive edge of the dclk clock signal, if the reset RST_N is high, then if the dack and dreq is low at the destination domain, dack must be low at the next cycle.

- **NRQ_WO_DACK:**

- `@(posedge u_Handshake_syn.sclk) disable iff (~u_Handshake_syn.rst_n) (!(u_Handshake_syn.sreq) && u_Handshake_syn.sack) | => !(u_Handshake_syn.sreq)`
- At every positive edge of the sclk clock signal, if the reset RST_N is high, and if the sreq is low and sack is high at the source domain, then sreq must be low at the next cycle.

Handshake

- **REQ_NO_HOLD:**
 - `@(posedge u_Handshake_syn.sclk) disable iff (~u_Handshake_syn.rst_n) (u_Handshake_syn.sreq && !(u_Handshake_syn.sack)) | => (u_Handshake_syn.sreq)`
 - At every positive edge of the sclk clock signal, if the reset RST_N is high, then if the sreq is high and sack is low at the source domain, sreq must be high at the next cycle.

FIFO

- **POP_ON_EMPTY:**

- `@(posedge u_FIFO_syn.rclk) u_FIFO_syn.rinc |-> !(u_FIFO_syn.empty)`
- At every positive edge of the rclk clock signal, if rinc is high, then empty should be low at that same clock cycle.

- **PSH_ON_FULL:**

- `@(posedge u_FIFO_syn.wclk) u_FIFO_syn.winc |-> !(u_FIFO_syn.wfull)`
- At every positive edge of the wclk clock signal, if winc is high, then wfull should be low at the same clock cycle.

FIFO

- **RPT_NO_GRAY:**

- `@(posedge u_FIFO_syn.rclk) disable iff (~u_FIFO_syn.rst_n) ##1 $changed(u_FIFO_syn.rptr) |-> ($onehot(u_FIFO_syn.rptr ^ $past(u_FIFO_syn.rptr)))`
- At every positive edge of the rclk clock signal, if the reset rst_n is high, if the rptr changes value in the next clock cycle, then exactly one bit must change in rptr (The result of performing a bitwise XOR between the value of the rptr in the next clock cycle and its value in the current clock cycle must yield a one-hot encoded value).

- **WPT_NO_GRAY:**

- `@(posedge u_FIFO_syn.wclk) disable iff (~u_FIFO_syn.rst_n) ##1 $changed(u_FIFO_syn.wptr) |-> ($onehot(u_FIFO_syn.wptr ^ $past(u_FIFO_syn.wptr)))`
- At every positive edge of the wclk clock signal, if the reset rst_n is high, if the wptr changes value in the next clock cycle, then exactly one bit must change in wptr

After Functional Analysis

- You should pass all check for Functional / Metastability.

CDC Phases

Pairs Schemes Convergence Functional Metastability

All Checks

- clk2 -> clk3
- clk3 -> clk2
- clk1 -> clk2
- clk2 -> clk1

	Scheme Property	Tag	Scheme	Scheme Type	Detection Type
✓	...O_syn.rdata_no_write_on_full	PSH_ON_FULL	u_FIFO_syn.rdata	FIFO	UserDefined
✓	...syn.rdata_no_read_on_empty	POP_ON_EMPTY	u_FIFO_syn.rdata	FIFO	UserDefined
✓	...O_syn.rdata_wptr_gray_coded	WPT_NO_GRAY	u_FIFO_syn.rdata	FIFO	UserDefined
✓	...O_syn.rdata_rptr_gray_coded	RPT_NO_GRAY	u_FIFO_syn.rdata	FIFO	UserDefined
✓	...dshake_syn.dreq_data_stable	CTL_NO_STBL	u_Handshake_syn.dreq	NDFF	Automatic
✓	...dshake_syn.sack_data_stable	CTL_NO_STBL	u_Handshake_syn.sack	NDFF	Automatic
✓	...ndshake_syn.din_src_req_hold	REQ_NO_HOLD	u_Handshake_syn.din	Handshake	UserDefined
✓	...ndshake_syn.din_src_new_req	NRQ_WO_DACK	u_Handshake_syn.din	Handshake	UserDefined
✓	...shake_syn.din_dest_ack_hold	ACK_WO_SREQ	u_Handshake_syn.din	Handshake	UserDefined
✓	...shake_syn.din_dest_new_ack	NAK_WO_SREQ	u_Handshake_syn.din	Handshake	UserDefined
✓	...e_syn.din_data_stability_dest	DAT_HS_STBL	u_Handshake_syn.din	Handshake	UserDefined

Total: 11 Filtered: 11

CDC Configuration CDC Phases Waivers

CDC Phases

Pairs Schemes Convergence Functional Metastability

All Tasks

- <embedded> 0:0:0:0
- <CDC_scheme_propert... 11:0:0:0

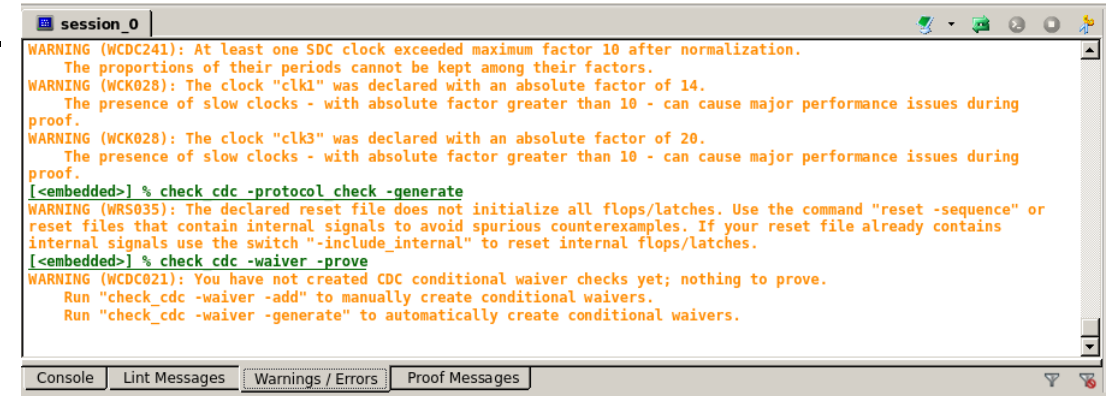
Type	Name	Engine	Bound
✓ Assert	CDC_u_FIFO_syn.rdata_no_write_on_full	PRE	Infinite
✓ Assert	CDC_u_FIFO_syn.rdata_no_read_on_empty	PRE	Infinite
✓ Assert	CDC_u_FIFO_syn.rdata_wptr_gray_coded	Hp (13)	Infinite
✓ Assert	CDC_u_FIFO_syn.rdata_rptr_gray_coded	Hp (41)	Infinite
✓ Assert	CDC_u_Handshake_syn.dreq_data_stable	Hp (8)	Infinite
✓ Assert	CDC_u_Handshake_syn.sack_data_stable	Hp (62)	Infinite
✓ Assert	CDC_u_Handshake_syn.din_src_req_hold	Hp (15)	Infinite
✓ Assert	CDC_u_Handshake_syn.din_src_new_req	Hp (15)	Infinite
✓ Assert	CDC_u_Handshake_syn.din_dest_ack_hold	Hp (7)	Infinite
✓ Assert	CDC_u_Handshake_syn.din_dest_new_ack	Hp (7)	Infinite
✓ Assert	CDC_u_Handshake_syn.din_data_stability_dest	Hp (103)	Infinite

Total: 11 Filtered: 11 Selected: 0 Validity: 11:0:0:0 Run: 0:0:0:11 Total: 22 Filtered: 0

CDC Configuration CDC Phases Waivers

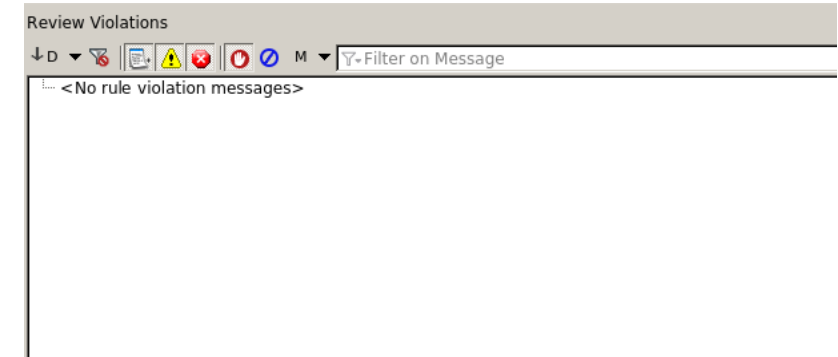
Summary

☐ There should be no error message in console.



```
session_0
WARNING (WDC241): At least one SDC clock exceeded maximum factor 10 after normalization.
The proportions of their periods cannot be kept among their factors.
WARNING (WCK028): The clock "clk1" was declared with an absolute factor of 14.
The presence of slow clocks - with absolute factor greater than 10 - can cause major performance issues during
proof.
WARNING (WCK028): The clock "clk3" was declared with an absolute factor of 20.
The presence of slow clocks - with absolute factor greater than 10 - can cause major performance issues during
proof.
[<embedded>] % check_cdc -protocol check -generate
WARNING (WRS035): The declared reset file does not initialize all flops/latches. Use the command "reset -sequence" or
reset files that contain internal signals to avoid spurious counterexamples. If your reset file already contains
internal signals use the switch "-include_internal" to reset internal flops/latches.
[<embedded>] % check_cdc -waiver -prove
WARNING (WDC021): You have not created CDC conditional waiver checks yet; nothing to prove.
Run "check_cdc -waiver -add" to manually create conditional waivers.
Run "check_cdc -waiver -generate" to automatically create conditional waivers.
```

☐ No violation message



Violation Key	Violation Type	Check	Tag	Severity	Rule Value	Design Value	Failure Reason	# Pairs	Source Reset Domain	Destination Re

☐ Nothing in violations.csv